

Research Methodology

Simulations

Peter Gallert

Department of Computer Science
School of Information Technology
Faculty of Computing and Informatics
Namibia University of Science and Technology

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Outline

- 1 Introduction
- 2 Example 1: Packet Tracer
- 3 Example 2: ns-3



Brain in a jar

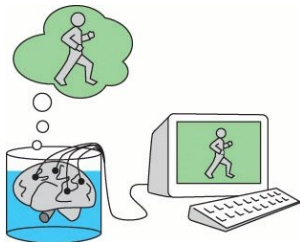


Figure: “Brain in a vat”. Picture by Alexander Wivel, CC-BY-SA

- If you were just a brain in a jar
- and had no sensory organs whatsoever
- with some computer to give input
- but no way to experience reality
- There would be no way to figure that out! (contrary to *The Matrix* Trilogy)
- A.k.a a **simulation**

What is a simulation?

- A **simulation** is a simplified imitation of a real-world process over time
- Simulations use a **model** of perceived reality
- Some characteristics are deemed important, others are not
- By using the simulation (the model), we **abstract from** all unimportant details
- Example: flight simulator
 - **Important:** pilot input, weather, airport, airplane type, mechanics, emergencies, ...
 - **Unimportant:** pilot mental health, airplane specimen, passengers, ticket price, ...
- Different from an **emulation** which has **no model** and attempts **no abstraction**, for instance VMWare



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Why simulations?

1 Money:

- Cisco "pod" (3 routers, 3 switches, and accessories): 100,000 N\$
- Cisco Packet Tracer: 0 N\$
- And that's still reasonably cheap! There are simulations for power grids, satellite launches, and the global financial system

2 Safety: How do you learn to

- Land a Boeing 747?
- Transplant a human heart?
- React to a pandemic?
- Reanimate an accident victim?
- Extinguish a fire in a residential building?

3 Convenience:

- Book lab 5, travel to campus, do a networking lab,
- Or: Fill in a set of forms, get HoD approval, take a Cisco pod home,
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Simulations in science

- Empirical scientific method: **experiment**
- Experiment is only valid if it is **repeatable**
- Many factors hinder repeatability:
 - The person / team running the experiment
 - Environmental factors: ambient temperature, time of day / year
 - Equipment manufacturing faults and deviations
 - Detail of the documentation
- A **simulation** helps suppress outside factors (abstraction)
- Also simplifies documentation
- Some simulations are **better than the real thing!**



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Packet Tracer



- **Packet Tracer** is a network simulator for teaching
- Developed by Cisco Systems, a leading networking hardware manufacturer
- **Real-time simulator** for routers, switches, firewalls, cell phones, access points, et cetera
- GUI, drop devices on screen, connect, configure
- Simulation mode: "follow" a packet on its way through the network
- Simple download, all operating systems, free of charge (account creation at www.netacad.com required)



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Main screen

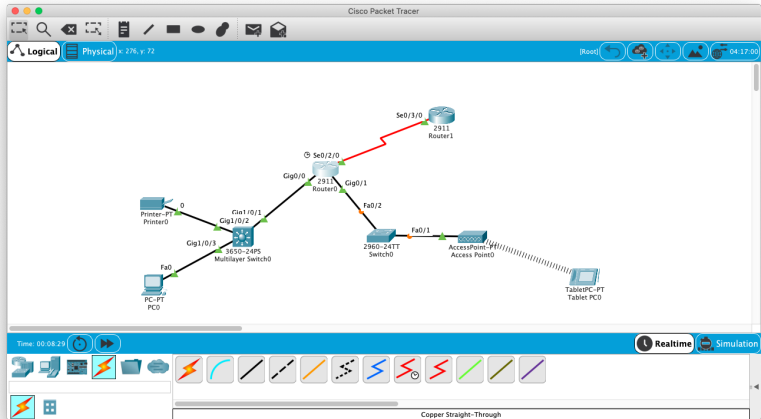


Figure: Packet Tracer screenshot with some connected devices. Fair use. 

Packet Tracer properties

- **Modelled:** Device configuration (hardware and software), protocol behaviour and performance
- **Not modelled:** Space considerations, reduced feature set
- Only selected commands and hardware options—those taught in, and sold to, the Cisco Academies



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Packet Tracer hardware configuration

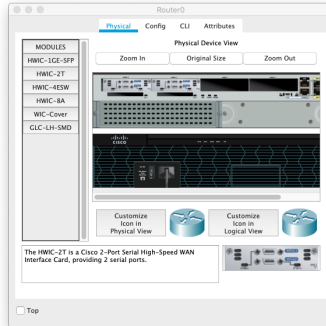


Figure: Packet Tracer configuration window: hardware. Fair use.



Packet Tracer software configuration

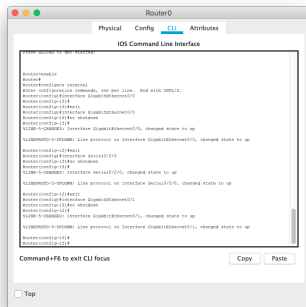


Figure: Packet Tracer configuration window: Cisco IOS command line interface. Fair use.

Advantages, disadvantages

- **Good for teaching computer networking:**

- 1 Point-and-click, easy to learn
- 2 Realistic simulation of hardware installation and configuration mistakes
- 3 Runs on all major operating systems, even on Android
- 4 Accompanied by an online course (CCNA) to teach network device configuration

- **Not suitable for performance estimates:**

- 1 Timers independent of host performance
- 2 Illogical result, for instance for Quality of Service impact

- **Proprietary and vendor-dependent:**

- 1 Non-Cisco devices cannot be simulated
- 2 Many "fantasy devices" that do not exist in this form



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ns-3



- **Network simulator (ns)** is a network simulator for research
- Currently in version 3, **ns-3**
- Developed by its user community worldwide
- **Discrete event simulator**, calculating **events**, their **effects**, **duration**, and **output**
- **Input:** C++ source code
- **Output:** (all optional)
 - .pcap packet capture file for display with Wireshark
 - .tr bare text trace file of chronological events
 - .xml graphic for display with offline animator NetAnim
 - Command line messages



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Command line

```

peter@debian: ~/work/ns-3-allinone/ns-3-dev
File Edit View Search Terminal Help
peter@debian:~/work/ns-3-allinone/ns-3-dev$ ./waf --run stest
Waf: Entering directory `/home/peter/work/ns-3-allinone/ns-3-dev/build'
[2809/2857] Compiling scratch/stest.cc
[2818/2857] Linking build/scratch/stest
Waf: Leaving directory `/home/peter/work/ns-3-allinone/ns-3-dev/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (5.047s)
Total Bytes Received: 680448
peter@debian:~/work/ns-3-allinone/ns-3-dev$ ./waf --run stest
Waf: Entering directory `/home/peter/work/ns-3-allinone/ns-3-dev/build'
[2810/2857] Compiling scratch/stest.cc
[2818/2857] Linking build/scratch/stest
Waf: Leaving directory `/home/peter/work/ns-3-allinone/ns-3-dev/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (2.570s)
At time +2s client sent 1024 bytes to 10.1.2.4 port 55
At time +2.01799s server received 1024 bytes from 10.1.3.3 port 49153
At time +2.01799s server sent 1024 bytes to 10.1.3.3 port 49153
At time +2.03371s client received 1024 bytes from 10.1.2.4 port 55
At time +3s client sent 1024 bytes to 10.1.2.4 port 55
At time +3.00529s server received 1024 bytes from 10.1.3.3 port 49153
At time +3.00529s server sent 1024 bytes to 10.1.3.3 port 49153
At time +3.01058s client received 1024 bytes from 10.1.2.4 port 55
peter@debian:~/work/ns-3-allinone/ns-3-dev$

```

Figure: Command line screenshot of ns-3, with simulation output. Own work



ns-3 properties

- **Modelled:** Protocol behaviour and performance
- **Not modelled:** most device configuration steps, hardware restrictions
- With ns-3, all devices **behave exactly as they should:**
 - Implementing all aspects of the simulated protocols
 - Delivering exactly the performance as determined by standards and configuration
 - Only runtime differs
- But many default values make little sense, for instance Ethernet with 4.2 GBps and no delay



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ns-3 C++ script

```

// Step 4: Define global declarations

GlobalValue::Bind ("ChecksumEnabled", BooleanValue (true));
// Calculate FCSs

// Step 5: Create nodes and node containers

NodeContainer routers;
routers.Create (2);

NodeContainer whkAP;
whkAP.Add (routers.Get (0)); // attach AP to router Windhoek
NodeContainer whkSTA;
whkSTA.Create (12);

NodeContainer rehoLAN;
rehoLAN.Add (routers.Get (1)); // attach switch to router Rehoboth
rehoLAN.Create (8);

// Step 6: Create connections

PointToPointHelper whkreho; // Define parameters of the Windhoek-Rehoboth WAN
whkreho.SetDeviceAttribute ("DataRate", StringValue ("5Mbps"));
whkreho.SetChannelAttribute ("Delay", StringValue ("2ms"));

CsmaHelper rehoswitch; // Define parameters of the Rehoboth LAN
rehoswitch.SetChannelAttribute ("DataRate", StringValue ("100Mbps"));
rehoswitch.SetChannelAttribute ("Delay", TimeValue (NanoSeconds (6560)));

Ssid whkssid = Ssid ("Windhoek WiFi");
// Create physical layer instance for Windhoek WiFi

```

Figure: C++ source code to produce a simulation. Own work

Advantages, disadvantages

- **Ideal for scientific experiments:**

- 1 The same C++ script will produce the same output, on any computer. Only runtime can differ
- 2 Control variables do not change
- 3 Easy control of the dependent variable, easy re-run with different values. For instance, same data transfer with 802.11a, 11n, 11ax
- 4 Manufacturing faults or marketing tricks irrelevant

- **Steep learning curve:**

- 1 Requires Linux, and only Linux
- 2 Tricky installation, lots of prerequisites, must be compiled on target machine
- 3 Lots of networking knowledge needed: Any logical mistake in the C++ script breaks compilation
- 4 Some understanding of C++ required—classes, inheritance, screen output, and general syntax



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NetAnim

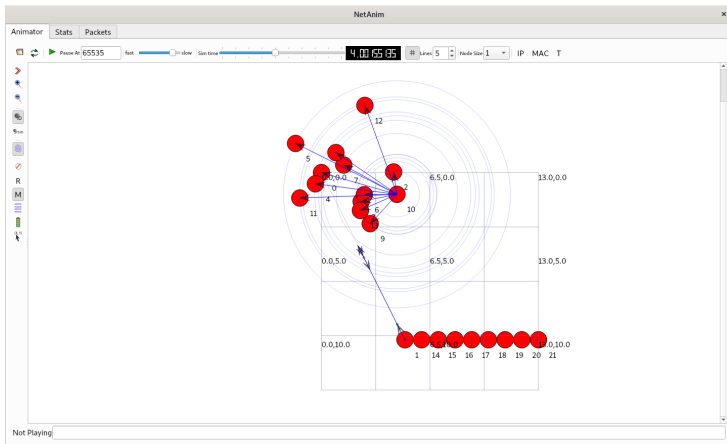


Figure: Graphical display of an ns-3 simulation with NetAnim. Own work



End of Presentation

Any questions?

